

Department of Computer Science

Course: AI Programming

Lab Number: Lab 4: Functions & Scope in Python

1. Objective

- By the end of this lab, students should be able to:
- Define and use **functions**
- Understand **local vs global variables**
- Differentiate between **return and print**
- Apply **modularity and code reuse**

1. What is the output of the following codes?

<pre>def double(y): x = 2 return x * y res = double(5) print(res)</pre>	
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<pre>x = 10 def test(): x = 5 print(x) test() print(x)</pre>	
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<pre>x = 10 def test(): global x x = 5 test() print(x)</pre>	
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2. Write a function `square(x)`:
 - Takes a number
 - Returns its square
3. Write a function `sumToN(n)`:
 - Returns the sum of numbers from 1 to n. (*Hint: similar to lecture example*)
4. Write a program that includes:
 - Function `cube(x)` $\rightarrow$ returns x^3
 - Function `avg(a, b, c)` $\rightarrow$ returns the average